

project2_py\local_minimizers.py

```
1 import numpy as np
2
3 def adam_optimizer(f, g, x0, n, count, alpha, gamma, gamma_v, gamma_s, epsilon, k=0, vel=None, sqr=None,
cost_g=None):
4     path = [x0]
5     x_best = x0
6     x_len = len(x0)
7     if sqr is None: sqr = np.zeros(x_len)
8     if vel is None: vel = np.zeros(x_len)
9     if cost_g is None: cost_g = 3 + x_len
10
11     while count() + cost_g <= n:
12         k += 1
13         gradient = g(x_best) # Costs
14         vel = gamma_v*vel + (1-gamma_v)*gradient
15         sqr = gamma_s*sqr + (1-gamma_s)*(gradient**2)
16         vel_hat = vel / (1 - gamma_v**k)
17         sqr_hat = sqr / (1 - gamma_s**k)
18         step = alpha * vel_hat / (epsilon + np.sqrt(sqr_hat))
19         x_best = x_best - step
20         path.append(x_best)
21         alpha *= gamma
22         if np.linalg.norm(step) < epsilon:
23             break
24     return path, alpha, k, vel, sqr
```